



Chapter 2: National and Regional Trends and Forecasts

Demographic, Economic, and Land Use Trends and Socioeconomic Forecasts

1. Demographic and Economic Context

Understanding demographic shifts, economic trajectories, and development patterns is fundamental to forecasting future travel demand and planning transportation infrastructure. The Tri-Cities region—comprising the independent cities of Colonial Heights, Hopewell, and Petersburg, alongside portions of Chesterfield, Dinwiddie, and Prince George Counties—exhibits distinct socioeconomic trends that directly influence mobility requirements.

2. Density and Activity Distribution

Figure 2 shows population density across the Tri-Cities region by Census block group. High-density residential pockets are concentrated within the urban cores of Petersburg, Hopewell, and Colonial Heights, as well as suburban communities in northern Chesterfield County.

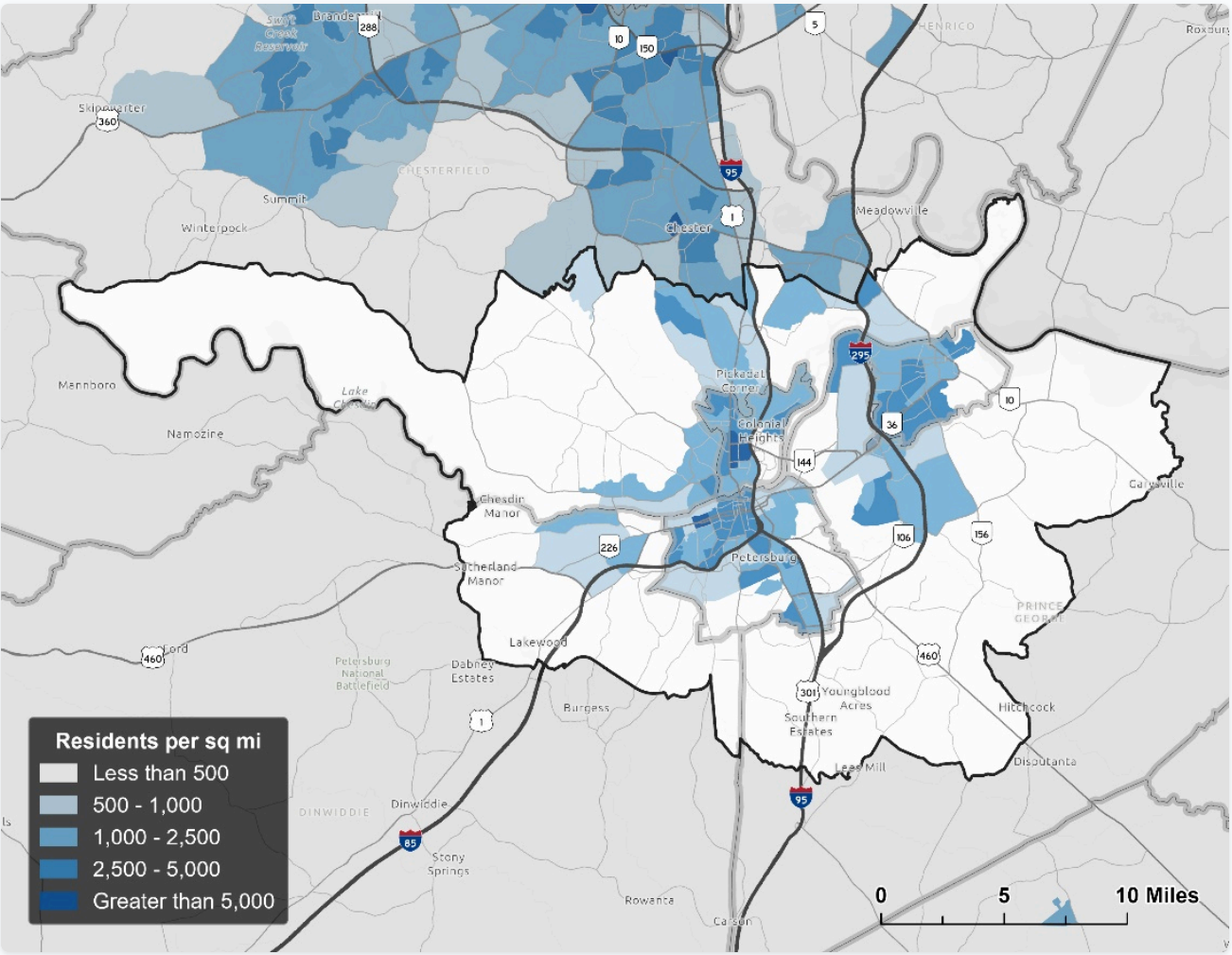


Figure 2. TCAMPO Population Density by Block Group (Residents per Square Mile)

Figure 3 illustrates the region's employment density. Commercial and industrial jobs are heavily concentrated along the I-95 and Route 1 corridors, Fort Gregg-Adams, the Port of Richmond/James River industrial zones in Hopewell, and retail centers in Colonial Heights.

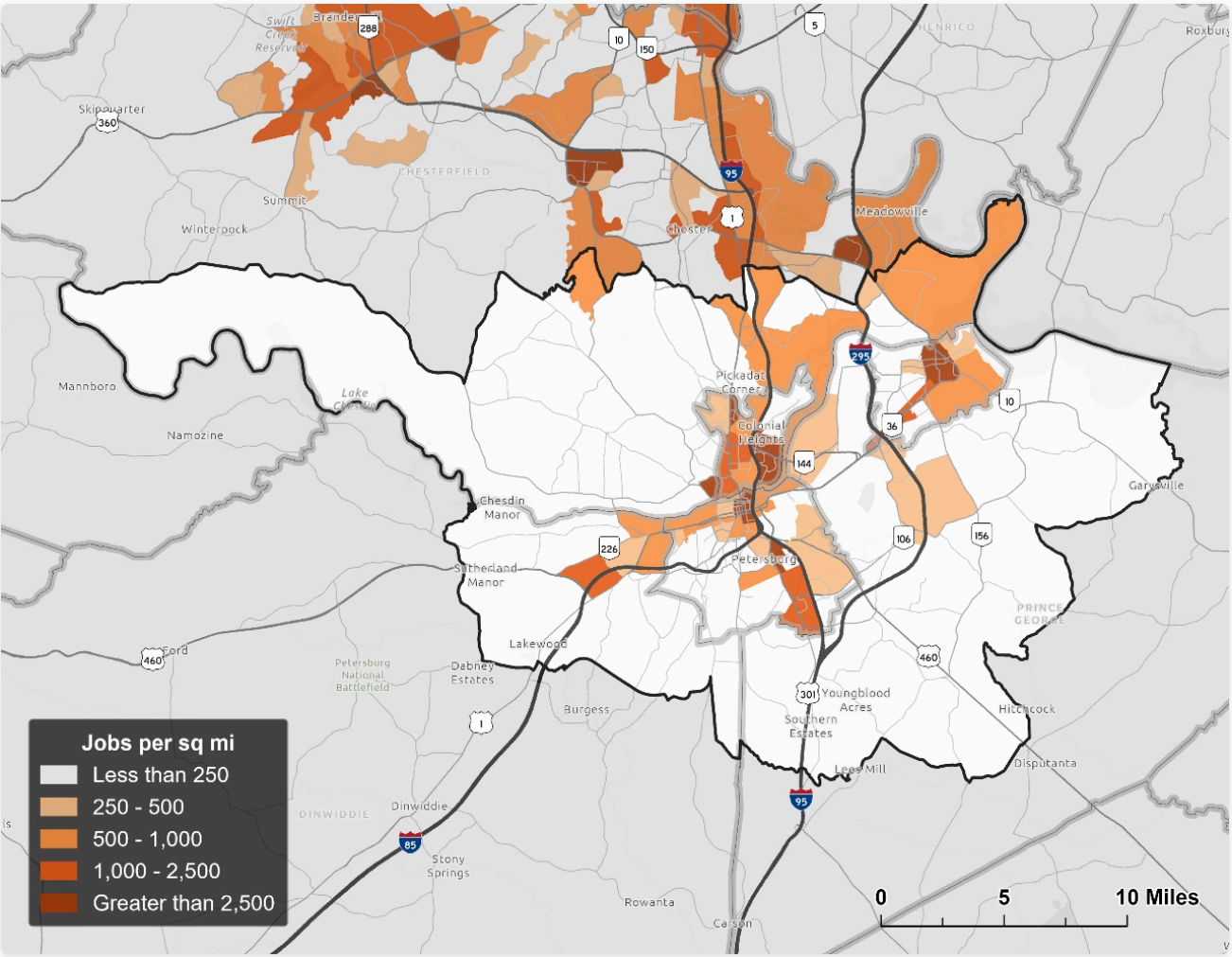


Figure 3. TCAMPO Employment Density by Block Group (Jobs per Square Mile)

Figure 4 combines population and employment density to illustrate activity density. This metric highlights prime corridors for multimodal investments, frequent transit service, and complete streets improvements.

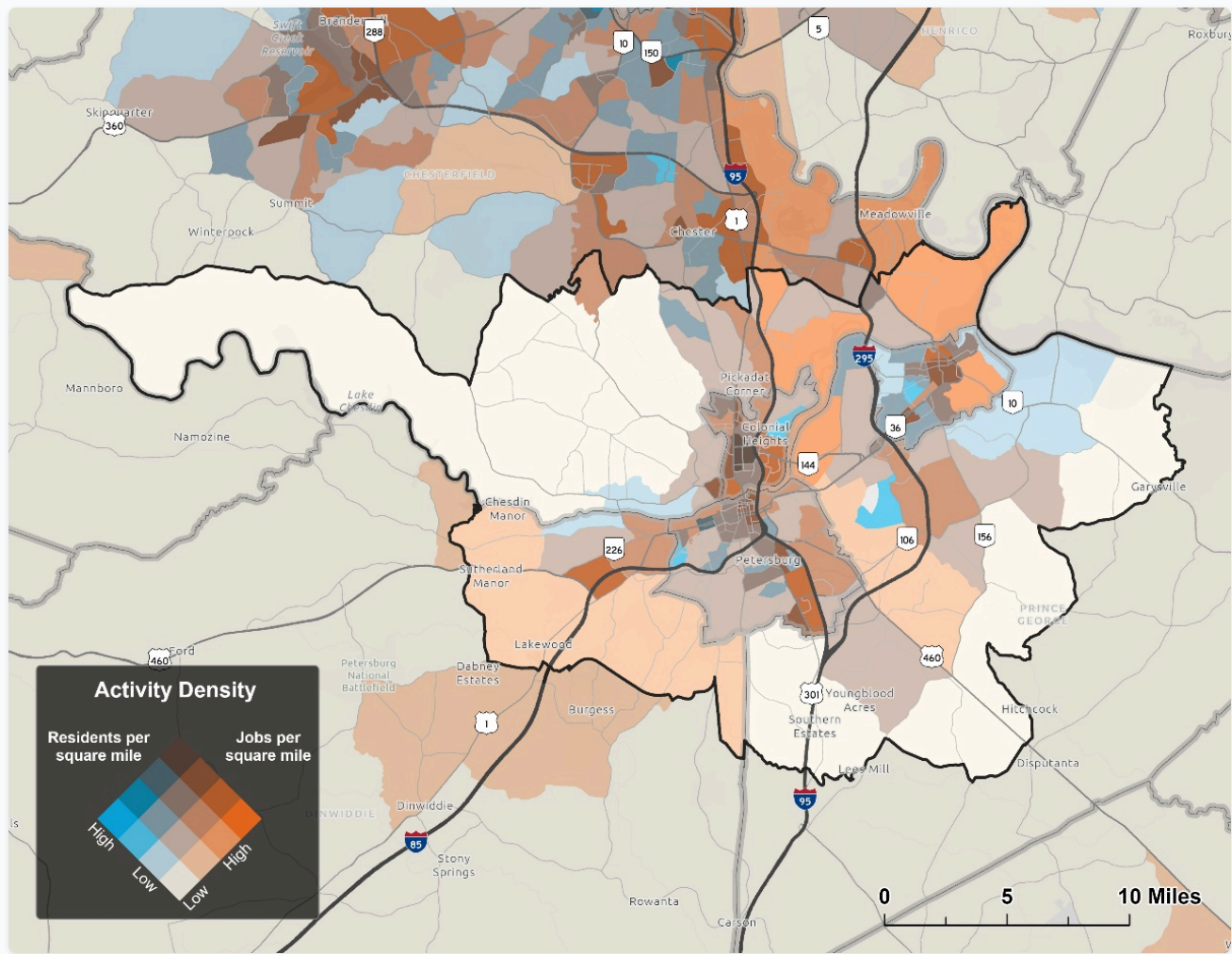


Figure 4. TCAMPO Activity Density by Block Group (Residents plus Jobs per Square Mile)

3. Population, Household, and Employment Trends

Between 1990 and 2020, TCAMPO's total population increased by 24.4%, growing from 134,978 to 167,920 residents. While substantial, this growth rate trailed the Commonwealth of Virginia (39.5%) and the United States (33.3%). Substantial variation exists across member jurisdictions: Chesterfield and Prince George counties experienced robust suburban and exurban growth, while the cities of Petersburg and Hopewell saw modest population contractions before stabilizing in the late 2010s.

Industry Composition and Economic Shift-Share

The regional economic base is anchored by federal government and defense (Fort Gregg-Adams), manufacturing and chemical processing (Hopewell industrial complex), healthcare, logistics, and retail trade. Shift-share analysis compares regional employment performance against national growth benchmarks across industry supersectors.

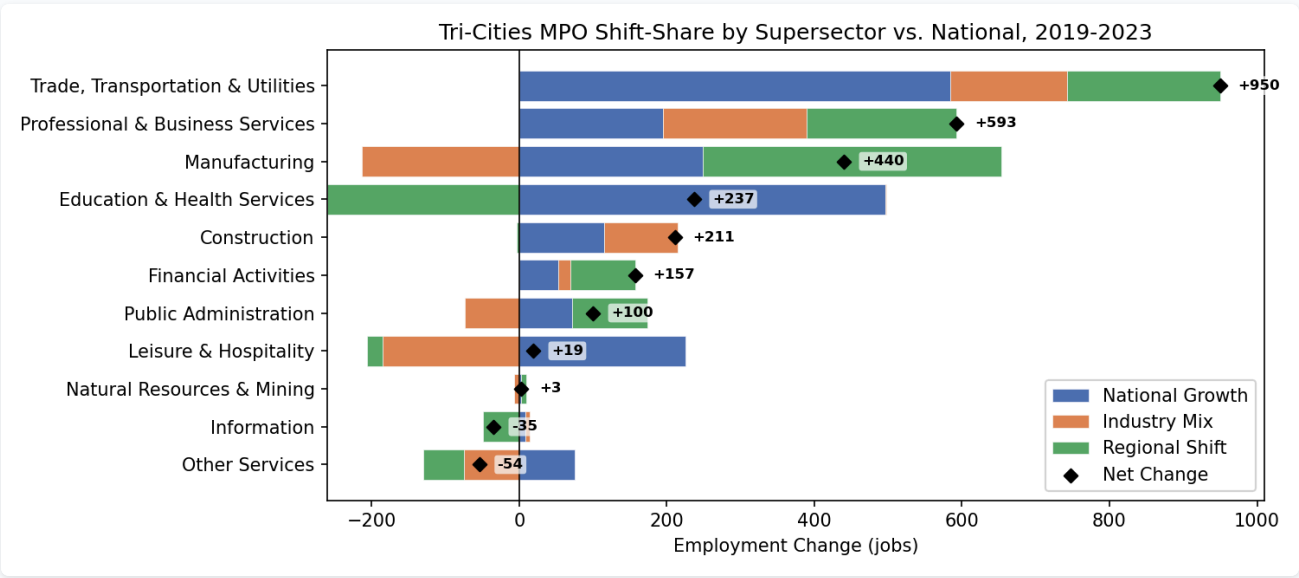


Figure 5. TCAMPO Shift-Share Employment Analysis by Supersector vs. National Benchmarks (2019–2023)

4. Demographic Profiles and Environmental Justice

In compliance with federal Title VI and Environmental Justice (EJ) mandates, TCAMPO identifies historically underserved communities, including minority, low-income, limited English proficiency (LEP), zero-vehicle households, elderly, and disabled populations. The Composite Environmental Justice Index identifies areas requiring focused public participation and equitable capital investments.

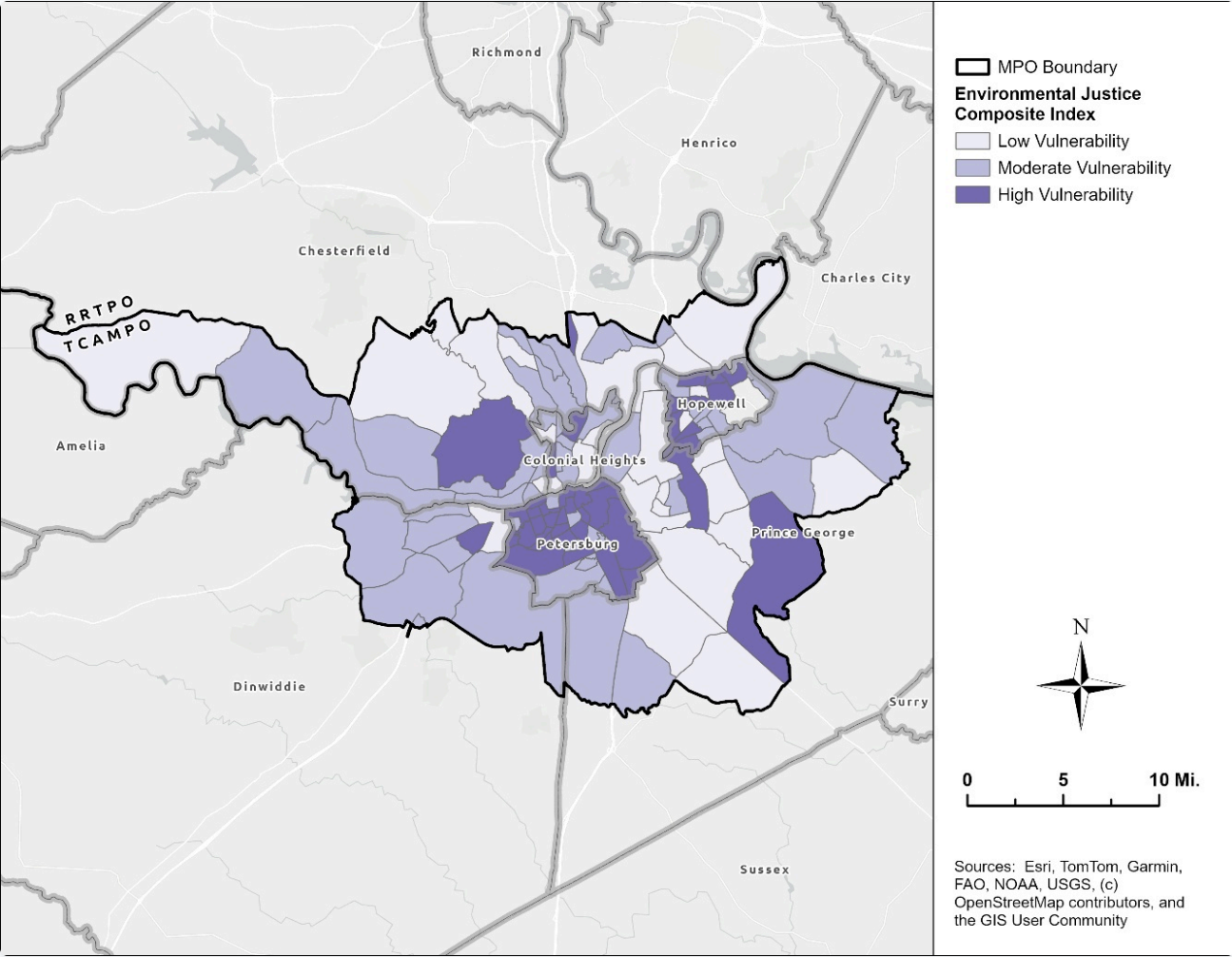


Figure 6. Composite Environmental Justice (EJ) Index by Census Block Group

5. Socioeconomic Forecasts (2017–2045)

Travel demand modeling requires projections of future population, households, and employment. Table 1 summarizes the adopted socioeconomic forecasts for the TCAMPO planning area.

Jurisdiction	2017 Population	2045 Population	Pop. Change (%)	2017 Employment	2045 Employment	Emp. Change (%)
Colonial Heights	17,731	18,920	+6.7%	12,410	14,100	+13.6%
Hopewell	22,644	24,150	+6.7%	10,230	11,800	+15.3%
Petersburg	31,644	34,500	+9.0%	15,920	18,450	+15.9%
Chesterfield (MPO Part)	47,210	64,800	+37.3%	18,340	25,600	+39.6%
Dinwiddie (MPO Part)	16,520	19,800	+19.9%	4,850	6,100	+25.8%
Prince George (MPO Part)	32,171	41,200	+28.1%	14,560	19,300	+32.6%
TCAMPO Total	167,920	203,370	+21.1%	76,310	95,350	+24.9%

6. Land Use and Development

Figure 7 and Figure 8 demonstrate historical and contemporary land cover transformations across the Tri-Cities. Over the past three decades, thousands of acres of agricultural and forested land have transitioned to low-to-medium density residential and commercial developments.

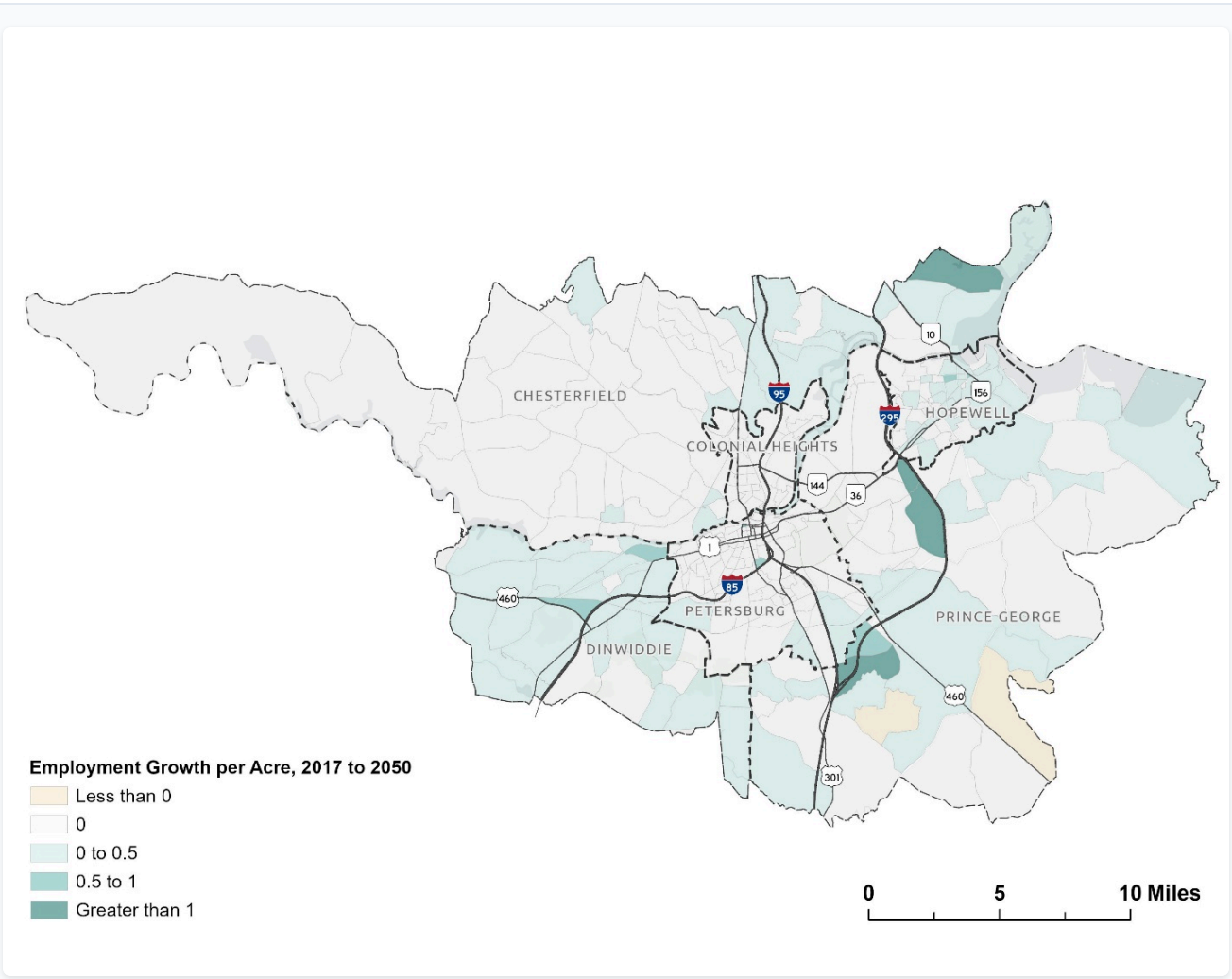


Figure 7. Existing Land Cover and Transitions from Undeveloped to Developed Areas

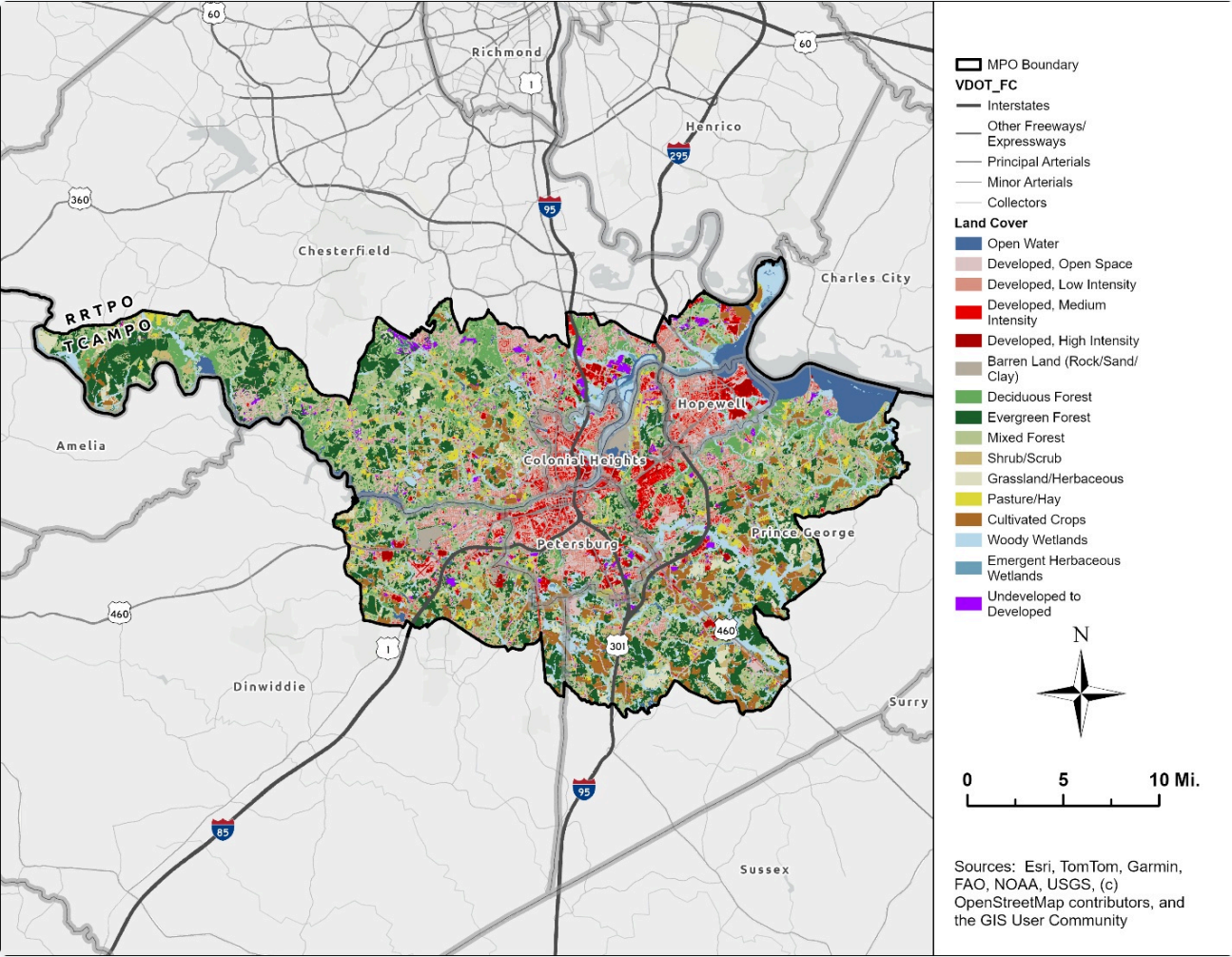


Figure 8. Percentage of Land that is Developed, Undeveloped, and Transitioned by Jurisdiction

7. Travel Patterns and Commuter Flows

Transportation demand is largely governed by journey-to-work dynamics and interregional travel. Table 2 details household vehicle availability, highlighting communities where transit and non-motorized alternatives are vital necessities.

Jurisdiction	Total Households	No Vehicle (%)	1 Vehicle (%)	2+ Vehicles (%)
Colonial Heights	7,340	6.1%	34.2%	59.7%
Hopewell	9,120	12.4%	39.8%	47.8%
Petersburg	13,580	18.7%	42.5%	38.8%
Chesterfield (MPO)	16,950	3.2%	26.4%	70.4%
Dinwiddie (MPO)	6,240	4.5%	28.1%	67.4%
Prince George (MPO)	11,480	4.1%	27.9%	68.0%
TCAMPO Average	64,710	8.4%	32.8%	58.8%

According to commuter flow analysis, over 42% of employed Tri-Cities residents commute daily outside the MPO boundary, primarily north into the Richmond metropolitan area along I-95, Route 1, and I-295. Conversely, major employment centers within the Tri-Cities draw significant inbound commuters from Southside Virginia and the Richmond suburbs, underscoring the critical need for interregional corridor coordination and transit connectivity.